VANET TRAFFIC CONGESTION DATASET ANALYSIS

1. Introduction

Vehicular Ad Hoc Networks (VANETs) enable connected vehicles to exchange information with other vehicles and roadside infrastructure. As road traffic becomes more congested, the communication channel can become busier and packets may experience greater delay and loss. This project investigates how four road-congestion states—Free-flow, Moderate, Heavy, and Gridlock—affect network communication performance.

2. Industry Scenario

A smart-transportation company operates a connected-vehicle communication system for traffic monitoring, safety messages, and intelligent transportation services. Network engineers need to understand whether increasing road congestion also degrades wireless communication. The analysis uses the supplied VANET traffic dataset and compares packet loss, communication delay, queue length, and channel busy ratio across the four congestion states.

3. Objectives

- Calculate the average packet loss percentage for each congestion state.
- Calculate the average communication delay for each congestion state.
- Calculate the average queue length for each congestion state.
- Calculate the average channel busy ratio for each congestion state.
- Compare the four congestion states using suitable charts.
- Identify the congestion state where communication performance becomes significantly worse.
- Recommend one practical network-level action to improve communication reliability.

4. Dataset Description

The supplied workbook contains 195,714 records and 27 columns. The congestion-state column is named 'label'. The four required states are Free-flow, Moderate, Heavy, and Gridlock. The communication variables used in this analysis are packet_loss_pct, avg_comm_delay_ms, queue_length_veh, and channel_busy_ratio_pct.

5. Methodology

1. Load the supplied vanet_traffic_data.xlsx file.
2. Group all records by the congestion label.
3. Calculate the arithmetic mean of each required communication metric for every congestion state.
4. Order the results from Free-flow to Gridlock to show the progression of congestion.
5. Create bar charts for direct comparison.
6. Interpret the point at which packet loss, delay, queue length, and channel utilization become substantially worse.

6. Number of Records by Congestion State

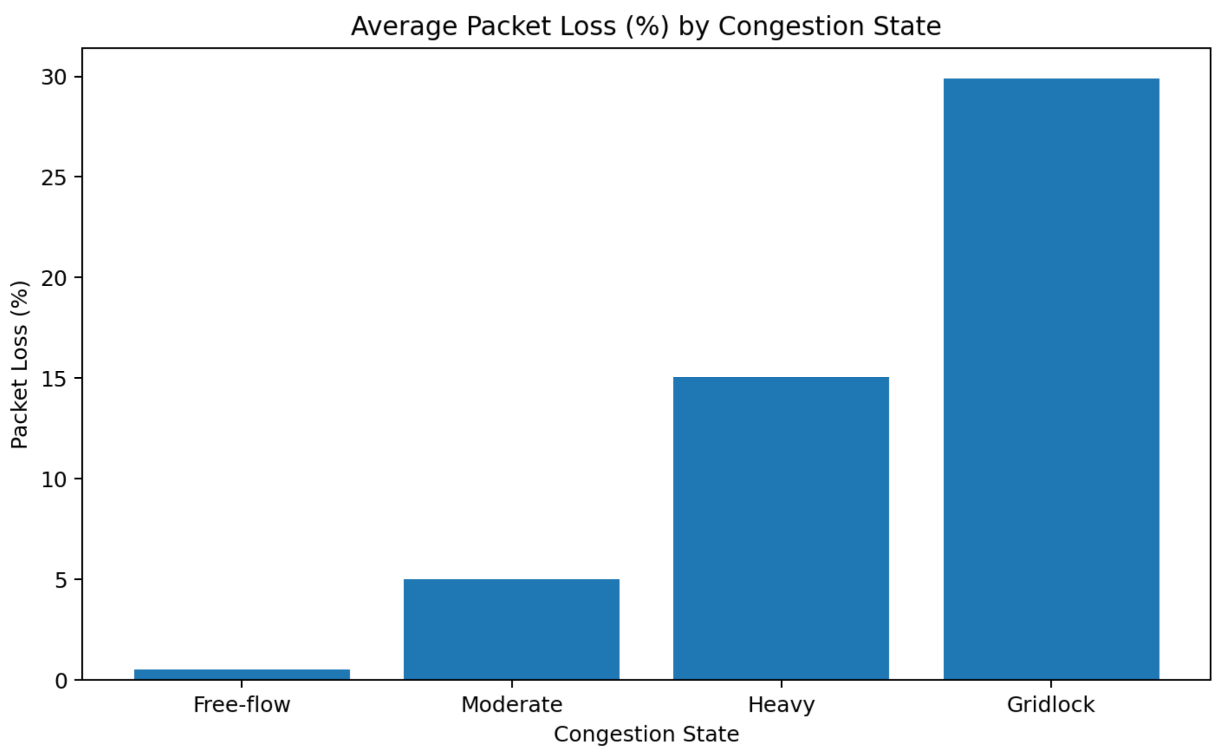
Congestion State	Records
Free-flow	57,812
Moderate	54,018
Heavy	47,440
Gridlock	36,444

7. Average Communication Performance

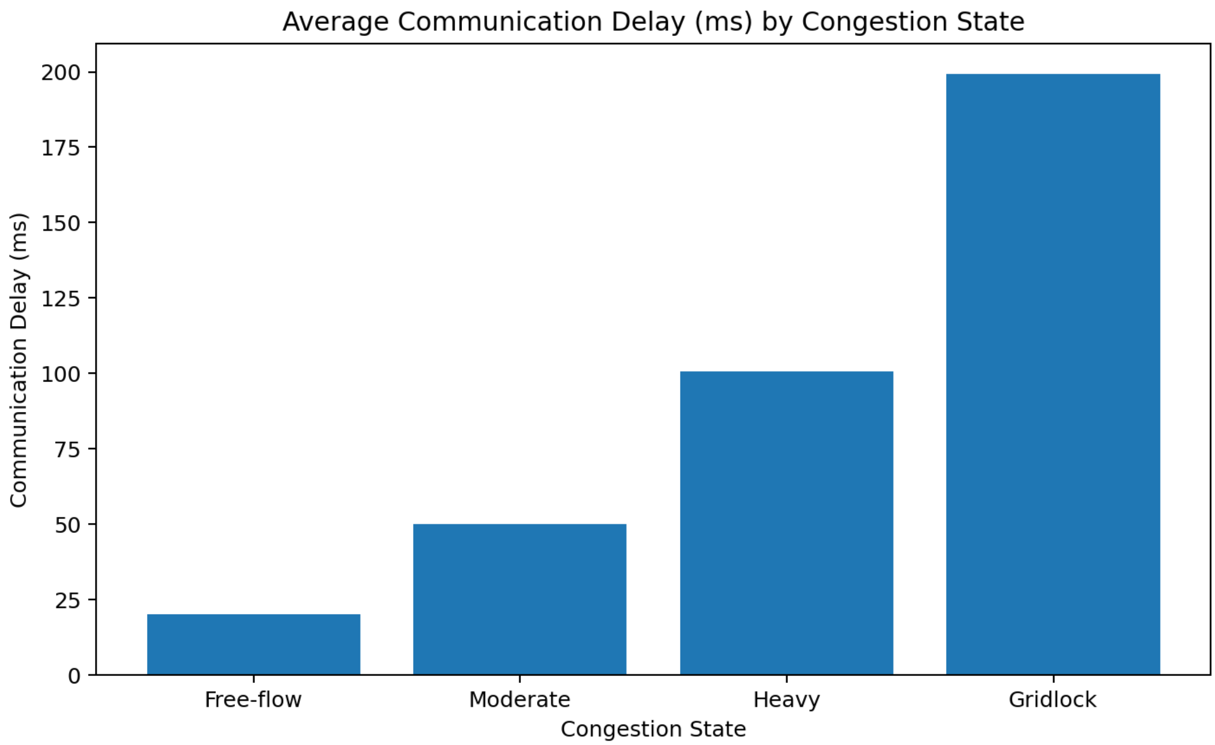
Congestion State	Packet Loss (%)	Delay (ms)	Queue Length	Channel Busy (%)
Free-flow	0.521	20.130	2.037	10.095
Moderate	5.007	49.945	9.991	29.935
Heavy	15.069	100.551	25.200	60.104
Gridlock	29.902	199.327	59.761	89.777

The averages show a clear deterioration in communication as congestion increases. Packet loss rises from 0.521% in Free-flow to 29.902% in Gridlock. Average communication delay rises from 20.130 ms to 199.327 ms. Average queue length increases from 2.037 to 59.761 vehicles, while channel busy ratio rises from 10.095% to 89.777%.

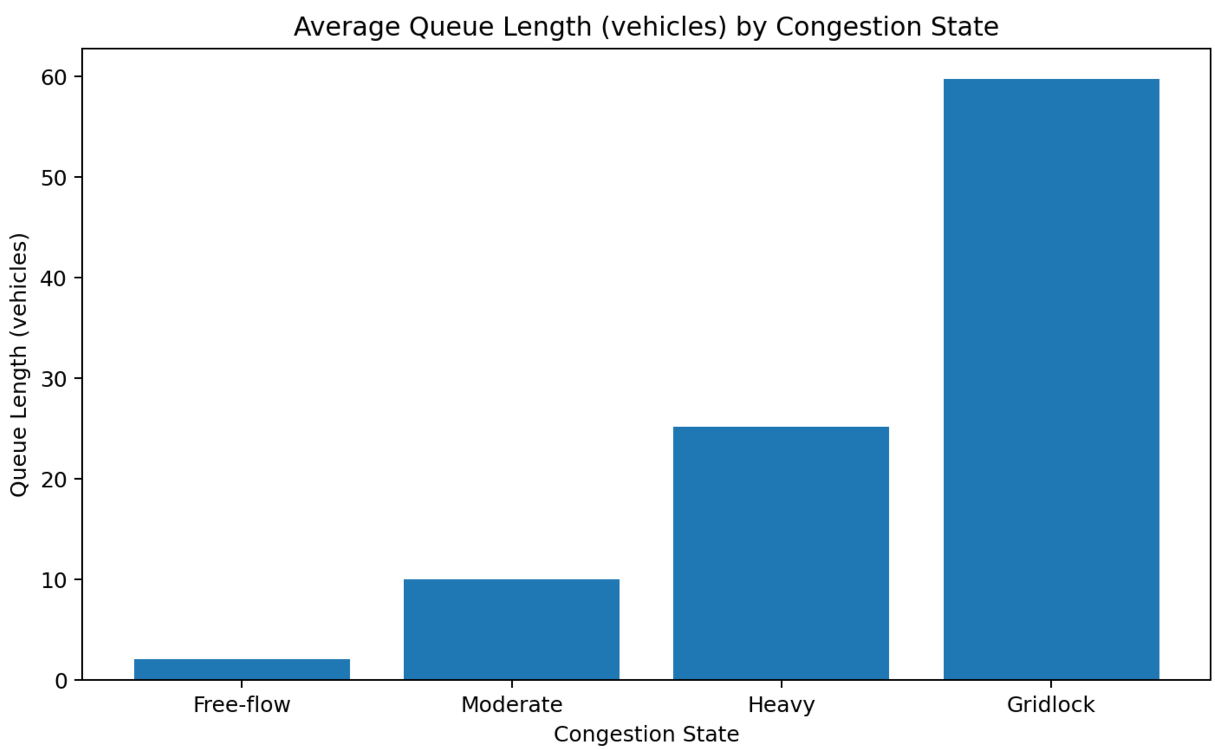
8. Bar Chart – Average Packet Loss



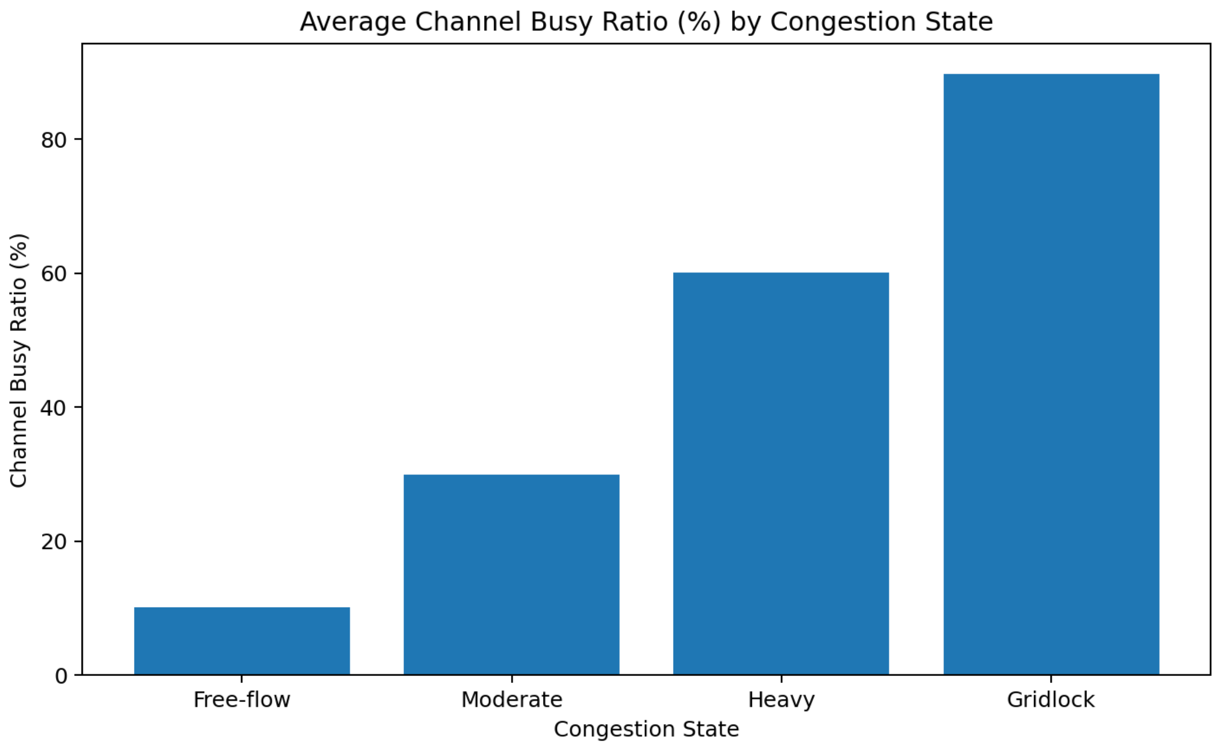
9. Bar Chart – Average Communication Delay



10. Bar Chart – Average Queue Length



11. Bar Chart – Average Channel Busy Ratio



12. Analysis and Interpretation

- Free-flow has the best communication performance, with only 0.521% packet loss and 20.130 ms average delay.
- Moderate congestion increases packet loss to 5.007% and delay to 49.945 ms.
- Heavy congestion causes a major deterioration: packet loss reaches 15.069%, delay reaches 100.551 ms, and channel busy ratio reaches 60.104%.
- Gridlock is the worst state. Packet loss reaches 29.902%, communication delay reaches 199.327 ms, queue length reaches 59.761 vehicles, and channel busy ratio reaches 89.777%.
- The transition from Moderate to Heavy is especially important because all four communication indicators increase sharply, indicating that network performance is becoming substantially worse before the system reaches Gridlock.

13. Percentage Increase from Free-flow to Gridlock

- Packet Loss (%): 5640.4% increase.
- Communication Delay (ms): 890.2% increase.
- Queue Length (vehicles): 2833.3% increase.
- Channel Busy Ratio (%): 789.4% increase.

14. Congestion State Where Performance Becomes Significantly Worse

The Heavy congestion state is the key deterioration point. Compared with Moderate congestion, average packet loss increases from 5.007% to 15.069%, communication delay increases from 49.945 ms to 100.551 ms, queue length increases from 9.991 to 25.200 vehicles, and channel busy ratio increases from

29.935% to 60.104%. This indicates that communication reliability and responsiveness are already seriously affected under Heavy congestion, with Gridlock representing the most severe condition.

15. Recommended Network-Level Action

Practical recommendation: implement congestion-aware priority scheduling with dedicated high-priority V2X safety-message queues. During Heavy and Gridlock conditions, safety and control messages should be given priority over lower-priority traffic, while adaptive channel access and message-rate control can reduce unnecessary channel occupancy. This can reduce packet collisions, limit queueing delay, and protect critical vehicle-to-vehicle and vehicle-to-infrastructure communication.

16. Conclusion

The analysis demonstrates a strong relationship between road congestion and VANET communication performance. As congestion progresses from Free-flow to Moderate, Heavy, and Gridlock, packet loss, communication delay, queue length, and channel busy ratio all increase. Gridlock has the poorest communication performance, while Heavy congestion is the critical point where degradation becomes substantial. A congestion-aware network priority mechanism that protects safety-critical V2X traffic is therefore recommended to improve communication reliability.

17. Digital Tools Used

- Microsoft Excel / Python-compatible spreadsheet processing for reading and organizing the supplied dataset.
- Python (Pandas) for grouping records and calculating averages.
- Matplotlib for creating bar charts.
- Microsoft Word-compatible DOCX format for the final project report.

18. Source

Source dataset: vanet_traffic_data.xlsx supplied for this project. Only the supplied dataset was used for the calculations and charts.